

A Branch A cohomological Higgs module: stabilizer-derived quartic overlap and lightness-protection gates

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May 19, 2026

Abstract

We describe a constrained Branch A projection/cohomology Higgs module in which the observed four-dimensional Higgs is treated as the visible projection of a tau-profiled parent field. The central reproducible calculation links a minimal $3 + 2$ stabilizer, the canonical hypercharge direction, a Higgs localization exponent $\nu_D = 3/10$, and a zero-mode quartic overlap $I_4(3/10) = 0.133756$. Under the stated Branch A rule, the observed Higgs quartic corresponds to an order-one parent quartic rather than to an extreme hierarchy. The construction remains a candidate mechanism with explicit validation gates. It is not a completed Standard Model derivation, not a proof of any parent projection theory, and not an empirical claim.

1 Scope and Claim Boundary

This manuscript isolates one theoretical module: a possible origin for a Higgs-scale quartic overlap and a possible cohomological lightness-protection route. The paper does not claim that the Standard Model, gravity, or a complete parent projection theory has been derived. The result should be read as a reproducible candidate mechanism with explicit proof obligations.

The closest conventional reference points are not grand-unified completion claims, but localization mechanisms. Domain-wall and defect-localized modes show that chiral or light fields can be tied to a wall profile [5, 6, 9]. Warped and extra-dimensional constructions show that overlap integrals can generate hierarchies without inserting every low-energy number directly [8]. BRST cohomology provides the standard language for separating physical representatives from quotient or gauge directions [1, 7]. The present module borrows none of these as evidence for Tau Core; instead, they identify the technical class of ideas against which the Branch A proposal should be judged: wall selection, normalizable zero modes, overlap-induced effective couplings, and cohomological protection.

2 Related Mathematical Structures

The proof gates above are deliberately aligned with established mathematical physics structures. The alignment is methodological rather than evidential: the cited frameworks show how similar technical problems are normally made well-defined, but they do not validate the Branch A module.

Branch A gate	Related structure	Use and non-claim
finite internal cell, compact spectrum	spectral triples and spectral action [3, 2, 4]	motivates the need for an algebra, Hilbert space, Dirac operator, compact spectrum, and index data; does not imply that the Branch A cell is a Connes geometry
wall profile and zero mode	domain-wall and defect localization [5, 9, 6]	supports the technical naturalness of wall-localized normalizable modes; does not derive the Branch A localization rule
overlap-generated couplings	extra-dimensional overlap mechanisms [8]	provides a familiar interpretation of effective couplings from profile overlaps; does not solve the Higgs hierarchy here
quotient/protection gate	BRST and cohomological control [1, 7]	supplies the language for quotienting unphysical directions and tracking anomalies; full Ward/anomaly closure remains open
C10–C13 uniqueness stack	index and cohomology logic [3, 4]	identifies the kind of theorem needed for protected residues and determinant pairing; no index theorem is proven in this manuscript

This comparison is meant to make the proof obligations sharper. The paper does not claim priority over, equivalence with, or derivation from these programs. It claims only that the Branch A candidate should be judged by similarly hard standards: explicit spectral data, admissible domains, index control, anomaly/Ward closure, and determinant finiteness.

The full Paper 4 reproducibility package, including the source-minimal repository, regeneration script, Wolfram audit scripts, compiled PDF, and arXiv source package, is archived at [doi:10.5281/zenodo.20285909](https://doi.org/10.5281/zenodo.20285909). The package can be regenerated with the commands listed in the repository README.

The detailed derivation ledger for the Branch A gates is distributed as a separate generated companion PDF in this repository. It collects the localization-forcing route, the ϵ_τ scale gate, the wall-cell embedding gate, the v0.2 single-package parent-action scaffold, and the dimensionless top/Yukawa coefficient chain. The present manuscript reports only the paper-level consequences and reproducible checks needed for the Higgs-module candidate.

3 Minimal Projection Setup

The tau coordinate is treated as an internal projection coordinate rather than an ordinary fifth spacetime dimension. A parent Higgs profile is written as

$$H(x^\mu, x), \tag{1}$$

where x^μ is ordinary four-dimensional spacetime and x is the internal tau/projection coordinate. The observed Higgs is the visible projection

$$\phi(x^\mu) = \Pi_{\text{vis}}[H] = \int h_D(x) H(x^\mu, x) dx. \quad (2)$$

Projection-null directions have the form $Q_D^\dagger \Lambda$ and are proposed to be quotiented out:

$$H \sim H + Q_D^\dagger \Lambda. \quad (3)$$

4 Branch A Stabilizer

Let the parent internal space split as

$$V \simeq \mathbb{C}^5 = C \oplus W, \quad \dim C = 3, \quad \dim W = 2. \quad (4)$$

The traceless two-block generator is

$$T_\Sigma = \frac{1}{\sqrt{60}} \text{diag}(2, 2, 2, -3, -3). \quad (5)$$

Its stabilizer is $S(U(3) \times U(2))$, giving the Standard-Model-compatible non-Abelian stabilizer structure. With

$$Y = \text{diag} \left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2} \right), \quad (6)$$

the canonically normalized hypercharge generator satisfies $T_\Sigma = -T_Y$. Equivalently,

$$T_Y = \sqrt{\frac{3}{5}} Y, \quad (7)$$

so the Branch A stabilizer fixes the familiar hypercharge-normalization factor $\kappa_\tau^2 = 3/5$ used below. This numerical factor is therefore not fitted to the Higgs quartic; it is inherited from the stabilizer-compatible normalization of the hypercharge direction.

Spectral origin of the normalization. The stronger version of this statement is not merely trace normalization. It would show that the compact tau geometry selects the same coefficient as a spectral invariant:

$$\kappa_\tau^2 = \frac{\text{Tr}_{\mathcal{F}}(T_Y^2)}{\text{Tr}_{\mathcal{F}}(Y^2)} = \frac{3}{5}, \quad (8)$$

with the trace, fiber, and normalization fixed by D_τ and the compact domain rather than by a chosen convention. Equivalently, the same value should appear as a heat-kernel/index residue of the protected hypercharge line:

$$a_Y(D_\tau) \propto \text{Res}_{s=0} \text{Tr}(T_Y^2 |D_\tau|^{-s}). \quad (9)$$

This is now a named gate: the spectral origin of κ_τ^2 must be derived from the compact geometry before the 3/10 exponent can be promoted from conditional target to theorem.

5 Higgs Exponent and Zero Mode

The Branch A localization postulate is treated here as an assumption/theorem-candidate:

$$\nu_i = \kappa_\tau^2 |Y_i| = \frac{3}{5} |Y_i|. \quad (10)$$

This separation is important. The factor $3/5$ follows from the Branch A hypercharge normalization above; the main theoretical blocker is the remaining localization postulate $\nu_i = \kappa_\tau^2 |Y_i|$. That postulate must eventually be derived from a stabilizer-compatible metric, a variational extremum, an index-theoretic constraint, anomaly matching, or another independent principle. Until such a derivation exists, the quartic-overlap result should not be read as a derivation of the Higgs quartic.

The limited claim tested here is narrower but meaningful. Among linear hypercharge-localization rules $\nu_i = c |Y_i|$, the value $c = 3/5$ is the Branch A working value that keeps the Higgs mode localized and gives an order-one parent quartic requirement. This establishes the quantitative target that the derivation gate must explain; it does not close that gate by itself.

The physical picture is simple. A wall-localized mode is not spread uniformly through the parent tau coordinate. It lives near the wall, and the amount of localization controls how much of the parent quartic survives into the visible four-dimensional quartic. In this manuscript the Higgs doublet is assigned the Branch A exponent $\nu_D = 3/10$. That number is not adjusted to the measured Higgs mass. It is the value obtained if the stabilizer-normalized hypercharge line supplies the localization charge. The proof obligation is therefore sharply defined: derive the wall connection that makes this assignment unavoidable.

Charge-to-exponent gate. The remaining nontrivial step is to derive the wall connection coefficient, not the zero-mode solution after the coefficient is given. The target theorem is

$$D_\tau \implies A_{\tau,i}(x) = \kappa_\tau^2 Y_i f(x), \quad f(x) = \tanh x, \quad (11)$$

with $\kappa_\tau^2 = 3/5$ fixed by the same canonical hypercharge normalization used in the stabilizer calculation. If this theorem closes, the Higgs doublet exponent is no longer an independent input:

$$\nu_D = \kappa_\tau^2 |Y_D| = \frac{3}{5} \cdot \frac{1}{2} = \frac{3}{10}. \quad (12)$$

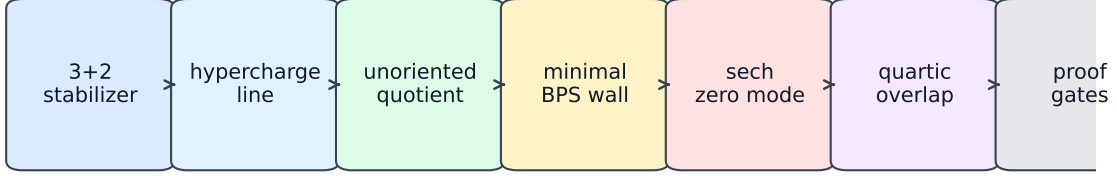
Thus the real open problem is the spectral/geometric origin of $A_{\tau,i}$; the value $3/10$ is then a consequence of the protected hypercharge eigenvalue and the universal wall metric coefficient.

6 Theorem Status Summary

Claim	Status in this manuscript
3 + 2 stabilizer	derived within the minimal Branch A carrier
two-block visible-sector selection	conditional parent/Hessian gate
minimal 3 + 2 carrier	conditional on two protected color/weak-support clusters
invariant roles ϵ_3, ϵ_2	conditional minimality gate
invariant/anomaly bridge	consistency audit; not representation derivation
$\kappa_\tau^2 = 3/5$ normalization	derived from canonical hypercharge normalization
unique Abelian direction	derived within the traceless 3 + 2 Branch A centralizer
unoriented T_Y line quotient	theorem-candidate, with projector audit
$\tanh x$ wall	candidate minimal/BPS wall route
$\nu_i = \kappa_\tau^2 Y_i $	theorem-candidate, not yet parent-derived
quartic overlap $I_4(3/10)$	computed and independently audited
projection-BRST protection	roadmap gate
localization rule $\nu_i = \kappa_\tau^2 Y_i $	conditional F1–F8 forcing route; full parent-action proof open
minimal forcing parent-action scaffold	quotient/stability/BPS/domain candidate; two-module origin and QFT closure open
top determinant / radiative stability	open gate
top/Yukawa slot and same-domain determinant pilot	sharpened open gate; not a physical determinant proof
single-package Branch A parent action v0.2	compact cell, wall domain, Q/regulator, and scale gate unified; not final action
Q-paired compact-spectrum toy	finite-dimensional same-domain Q demo; not physical orbifold theorem
orbifold Q-domain theorem	boundary-domain target stated; compact self-adjoint proof open
$y_t^{\text{parent}} = 0.9615522319$	fixed dimensionless overlap coefficient; not a top-mass prediction
ϵ_τ absolute scale	universal parent density/stress-energy gate; still open
Standard Model emergence program	separate roadmap, not a result of this paper

7 Parent-Selection Refinements

Recent parent-theory audits sharpen the status of the 3 + 2 input. They do not prove the parent action, but they reduce the apparent arbitrariness of the Branch A carrier. The current conditional



conditional route: derived pieces feed theorem-candidate gates before physical claims

Figure 1: Mechanism route and proof-gate structure. The first steps are controlled algebraic reductions; the localization rule and physical interpretation remain conditional on the parent-action gates.

chain is

$$\begin{aligned}
 &\text{two protected visible clusters} \rightarrow \text{minimal } 3 + 2 \\
 &\quad \rightarrow \epsilon_3 \text{ closure} + \epsilon_2 \text{ pairing} \\
 &\quad \rightarrow \text{anomaly-safe target gate.}
 \end{aligned} \tag{13}$$

The first step is a stability/cohomology requirement: the parent Hessian should produce exactly two protected visible light clusters, while any additional parent clusters must be Hessian-heavy or quotient-null. If an extra light visible cluster remains, the low-energy block-scalar centralizer gains extra freedom and the clean Branch A hypercharge line is no longer unique.

Given two visible clusters, the minimal carrier with one color-like support role and one weak-like support role is $3 + 2$. A sharper invariant reading is that the larger block is the first one with a three-index antisymmetric closure tensor ϵ_3 , while the smaller block is the first one with a two-index antisymmetric pairing tensor ϵ_2 . Larger blocks are not excluded by this manuscript, but they add tensor and centralizer freedom that would need a parent-derived penalty or quotient rule.

The invariant/anomaly bridge is also conditional. The declared $3 + 2$ target is anomaly-safe, while removing the 3×2 bifundamental bridge or adding an extra unpaired light doublet fails the anomaly/chirality gate. This shows which structures must be derived together. It does not derive the Standard Model representation content.

8 Post-Package Branch A Gate Refinements

The companion parent-theory hub now tracks several refinements that sharpen the open gates of this manuscript without closing them. They are included here to make the current status explicit,

not to expand the scope of the paper into a complete Standard Model derivation.

First, the physical first-order wall operator should not be introduced as an independent choice. The sharper gate is that the operator

$$Q_\mu = \partial_x + \mu \tanh x \quad (14)$$

must arise from the same wall-Hessian/domain package that selects the background wall and the visible zero mode. If Q_μ has to be chosen by hand after the wall is declared, the localization route remains underived.

Second, a finite same-domain wall- Q determinant pilot records the expected positive-mode pairing structure, but it is not a physical top determinant. The pilot is useful because it checks whether the determinant gate can be posed in the same operator domain as the Higgs zero mode. It does not compute the renormalized Standard Model top loop and does not establish radiative stability.

Third, the top-sensitive slot is now identified more sharply as the up-type closure channel schematically associated with

$$q_L - H - u_R^c. \quad (15)$$

This identifies where a top/Yukawa deformation would enter the Branch A structure. It does not derive the top mass, the top Yukawa strength, the family hierarchy, or the physical determinant.

Finally, the broader Branch A Standard Model emergence program is treated as a separate roadmap. This manuscript may supply one Higgs-module gate inside that larger program, but it does not claim to derive the full representation content, gauge dynamics, Yukawa sector, anomaly cancellation, or low-energy effective field theory.

9 Single-Package Parent-Action Update

The newest parent-hub refinement combines several previously separate gates into one v0.2 Branch A single-package scaffold. The proposed package has the schematic form

$$S_A^{(0.2)} = \epsilon_\tau [S_{\text{cell}} + S_{\text{wall}} + S_{\text{gap}} + S_Q + S_{\text{anom}} + S_{\text{prod}} + S_{\text{reg}}]. \quad (16)$$

Here S_{cell} selects a compact tau cell $\mathcal{K}_\tau$ and its measure/count sector, while the Branch A wall-Hessian domain $\mathcal{W}_A$ must be an induced subsector or quotient of that same cell:

$$\mathcal{W}_A \subseteq \mathcal{K}_\tau \quad \text{or} \quad \mathcal{W}_A = \mathcal{K}_\tau / \mathcal{H}_A. \quad (17)$$

The purpose of this update is not to claim the final Tau Core action. It is to prevent the Higgs wall, top/Yukawa overlap, regulator, and absolute scale from becoming disconnected adjustable pieces.

This is the central limitation of the present paper. The expression above is a structured action scaffold, not an explicit microscopic UV completion. A real parent theory would have to define the configuration space, the measure, the allowed fields, the symmetry principle, the continuum limit, and the renormalized low-energy map before the scaffold could be promoted to a physical action. In its current form the scaffold is useful because it forbids independent tuning of wall, regulator, and scale sectors; it is not yet the final parent dynamics.

The same package keeps the quotient-visible traceless field,

$$\Sigma = \Pi_{\text{tr}} \Phi, \quad \Pi_{\text{tr}}(X) = X - \frac{1}{5} \text{Tr}(X) I_5, \quad (18)$$

the unique wall channel $f(\Sigma) = 2\text{Tr}(\Sigma T_Y)$, the positive transverse/off-block gap terms, and the shared-boundary cohomology relations

$$QP = B, \quad QC = -B, \quad QB = 0. \quad (19)$$

The product and regulator sectors require the same $\mathcal{W}_A$ and the same Q package for wall, Hessian, Yukawa-overlap, and determinant operations.

The compact tau cell is therefore a blocker rather than a decorative detail. The next version must specify whether $\mathcal{K}_\tau$ is a compact manifold, a finite spectral triple, a graph/cell complex, a quotient Hilbert domain, or another mathematically controlled object. It must also define the topology, inner product, boundary conditions, spectrum, and admissible compactification mechanism. Without this, ϵ_τ and the wall-cell embedding remain named placeholders rather than derived physical structure.

10 Compact Tau Geometry As The Next Central Gate

The compact tau cell is the most important unresolved object in the present program. A credible next step would be to replace the symbolic label $\mathcal{K}_\tau$ by a concrete compact geometry with a declared operator algebra and spectral problem:

$$(\mathcal{K}_\tau, \mathcal{A}_\tau, \mathcal{H}_\tau, D_\tau, \mu_\tau). \quad (20)$$

Here $\mathcal{A}_\tau$ is the internal observable algebra, $\mathcal{H}_\tau$ is the parent Hilbert domain, D_τ is the parent spectral or Dirac-like operator, and μ_τ is the measure or trace functional. The Branch A wall would then have to appear as an induced quotient or defect sector of this same object, not as an independently declared one-dimensional coordinate:

$$\mathcal{W}_A = \text{Defect}(\mathcal{K}_\tau, D_\tau, T_Y) \quad \text{or} \quad \mathcal{W}_A = \mathcal{K}_\tau / \mathcal{H}_A. \quad (21)$$

This gives a concrete mathematical target. The compact geometry should select the $3 + 2$ stabilizer, produce the unoriented hypercharge line, generate the minimal wall profile, define the same-domain Q and regulator, and give a finite spectral density from which ϵ_τ can be computed. If those steps are obtained from the same $(\mathcal{K}_\tau, \mathcal{A}_\tau, \mathcal{H}_\tau, D_\tau, \mu_\tau)$ package, the current scaffold would become a genuine microscopic route. If they require separate choices, the framework remains a controlled but underived localization model.

The immediate calculational test is therefore not another fit to the Higgs mass. It is an existence and uniqueness problem: find a compact tau geometry whose low-lying Hessian/spectral data force

$$3 + 2, \quad T_Y, \quad f(x) = \tanh x, \quad Q_i = \partial_x + \kappa_\tau^2 Y_i \tanh x, \quad (22)$$

and whose regulated determinant is compatible with the BRST/anomaly and radiative-stability gates.

A first sharpened v0.1 target is now explicit enough to be falsifiable as a mathematical object:

$$\mathcal{K}_\tau = S^1/\mathbb{Z}_2, \quad \mathcal{A}_\tau = (C^\infty(\mathcal{K}_\tau) \otimes (M_3(\mathbb{C}) \oplus M_2(\mathbb{C})))^{\mathbb{Z}_2}, \quad (23)$$

$$\mathcal{H}_\tau = L^2(\mathcal{K}_\tau) \otimes (\mathbb{C}^3 \oplus \mathbb{C}^2) \otimes \mathbb{C}_{\text{wall}}^2, \quad D_\tau = \sigma_1 \partial_x + \sigma_2 \kappa_\tau^2 T_Y f(x) + D_{\text{gap}}. \quad (24)$$

This is not yet the final parent action. Its value is that the next theorem is now concrete: classify the self-adjoint orbifold domains and show whether this single package gives a protected zero-mode index, paired positive spectrum, and admissible zeta/heat-kernel regulator without adding a separate Higgs sector by hand.

The compact index target can be stated at the boundary level. On the fundamental interval $I = [0, L]$, a Q-compatible domain must make the boundary form vanish:

$$\langle Q_\nu \psi, \phi \rangle - \langle \psi, Q_\nu^\dagger \phi \rangle = [\psi^* \phi]_0^L = 0, \quad (25)$$

while preserving $Q_\nu : \mathcal{D}_- \rightarrow \mathcal{D}_+$ and $Q_\nu^\dagger : \mathcal{D}_+ \rightarrow \mathcal{D}_-$. Proving this for the physical orbifold domain is the bridge from the Q-first toy pairing to a real compact index theorem.

11 Detailed Gate Ledger And Companion Scope

The C0–C20 blocks are treated here as a compact gate ledger for the main paper. They are not meant to turn the manuscript into a complete microscopic derivation. Their role is to state which assumptions have been made sharper, which assumptions remain load-bearing, and which theorem-level steps would upgrade the Branch A Higgs module from a constrained scaffold to a stronger derivation.

The expanded derivational reading belongs to the companion full-derivation packet. The main paper should therefore be judged by the claim boundary, the reproducible overlap calculation, the sensitivity audit, and the explicit validation gates. The C-ledger records the route from compact tau geometry to localization and determinant protection without claiming that the final parent action has already been found.

Gate	Main role	Still open
C0 Candidate: Finite-Fiber Wall Spectral Cell; Adversarial Audit Of The C0 Candidate	A finite-fiber wall spectral cell links $\mathcal{K}_\tau = S^1/\mathbb{Z}_2$, the finite-fiber wall spectral cell, $3 + 2$, T_Y , $\tanh x$, Q_i , and the regulator route; the local factor includes $H_- = Q_\nu^\dagger Q_\nu$ and $\zeta_{D_\tau}(s)$.	The parent action must still select the finite fiber, the wall functional, the unoriented line, and the scale without endpoint fitting.
C1 Forcing Route For The Inserted Structures; C2 Closure Route For Projector, Determinant, And Scale	Two protected visible clusters and a Bogomolny/minimal-wall route are used to force the inserted $3 + 2$ and $\tanh$ structures; C2 closes the projector, determinant, and scale book-keeping.	The forcing route is conditional until derived from the final parent structure and a physical determinant.
C3 Explicit Compact Spectral Geometry Candidate; Adversarial Audit Of The C3 Candidate	A compact spectral geometry candidate makes the wall, fiber, gap, domain, and regulator live in one package.	The base, fiber, block split, gap, and wall action must be selected rather than declared.
C4 Selection-Functional Target	The compact selection functional I_{compact} penalizes compactness, gap, cohomology, anomaly, wall, and regulator failures; the target is that all six penalties vanish. In the expanded ledger this is written as the condition that all six penalties vanish.	No minimality theorem or competitor classification is yet proven.
C5 Local Spectrum Gate; C6 Compact Spectrum Boundary-Value Problem; C7 Competitor Audit For The C4 Minimum	The local Pöschl–Teller spectrum, compact boundary-value problem, and competitor audit define the first spectrum-level tests; the new Q-first toy demo shows exact positive-spectrum pairing when $H_\pm$ are built from one finite-dimensional Q .	A solved compact spectrum and uniqueness theorem are still missing.
C8 Minimality Theorem Candidate; C9 Compact Spectrum Pilot; C10 Uniqueness Proof-Obligation Split	The minimality theorem candidate, numerical compact spectrum pilot, and proof-obligation split turn the hidden assumptions into falsifiable subclaims.	The pilot is a sanity check only; it does not prove paired-spectrum closure or minimality.
C11 Algebraic Minimality Lemmas; C12 Protected-Form Selection Gate; C13 Cohomology/Index Selection Route; C14 Toy Complex Sanity Check	The algebraic and cohomological route asks whether protected 3-form and 2-form residues select the first finite survivor instead of arbitrary representation choice.	The toy complex does not yet constitute a full index theorem or anomaly-safe classification.
C15 Operator-Domain Ansatz; C16 Shared-Domain Compatibility Gate; C17 Self-Adjoint Domain Candidate	The wall, cohomology, index, determinant, and regulator are required to share one compact self-adjoint domain.	The self-adjoint extension and Ward/anomaly closure remain theorem-level requirements.
C18 Paired-Spectrum Verification Target; C19 Zeta-Determinant Cancellation Target; C20 Residue Classification Gate	The determinant route requires positive-mode pairing, zeta/heat-kernel cancellation, and a residue list restricted to kernel, index, anomaly, boundary, or frozen matching terms.	Radiative stability is not proven until all allowed residues are computed and forbidden finite counterterms are excluded.

This compression is deliberate. The full derivation packet preserves the longer C0–C20 ledger.

The main text keeps only the gate structure needed to understand why the remaining proof tasks are precise rather than rhetorical.

12 Localization-Forcing Update

The companion theory hub now separates two statements that were previously compressed into the localization postulate. The first is a conditional natural-origin statement:

$$\text{selected Branch A wall line} + \text{trace pairing} + \text{same-domain wall Hessian} \implies \nu_i = \kappa_\tau^2 |Y_i|. \quad (26)$$

The second is a stronger forcing-theorem candidate. It states sufficient parent-level requirements F1–F8 under which the wall/domain package itself is forced at leading order: trace quotient, two protected visible blocks, stabilizer-preserving block-scalar wall, one traceless Abelian wall line, universal trace pairing, unoriented two-vacuum quotient, minimal BPS factorization, and same-domain Hessian/ Q factorization.

Under those requirements the leading channel operator is

$$Q_i = \partial_x + \kappa_\tau^2 Y_i \tanh x, \quad (27)$$

and the normalizable kernel gives

$$\nu_i = \kappa_\tau^2 |Y_i|. \quad (28)$$

For the Higgs doublet, $|Y_H| = 1/2$ and $\kappa_\tau^2 = 3/5$, hence

$$\nu_H = \frac{3}{10}. \quad (29)$$

This is a significant sharpening of the earlier postulate, but it is not yet a completed parent-action proof. The compact action scaffold currently forces the trace quotient, trace pairing, leading BPS wall, and local localization consequence. It still leaves open the parent derivation of the two protected visible modules, the deeper origin of the unoriented-line quotient, and full BRST/anomaly/regulator/top closure.

13 Candidate Route To The Localization Rule

The localization postulate can be sharpened into a candidate derivation route. Assume that the tau-direction zero-mode operator for a field component i is a covariant first-order operator

$$Q_i = \partial_x + A_{\tau,i}(x), \quad (30)$$

and that the Branch A stabilizer induces a hypercharge-directed tau connection

$$A_{\tau,i}(x) = \kappa_\tau^2 Y_i \tanh x. \quad (31)$$

Then the chiral zero-mode equation

$$Q_i h_i = 0 \quad (32)$$

has the local solution

$$h_i(x) \propto \exp \left[- \int^x \kappa_\tau^2 Y_i \tanh u \, du \right] = \text{sech}^{\kappa_\tau^2 Y_i} x. \quad (33)$$

Normalizability selects the sign/chirality of the localized branch, so the positive localization exponent is

$$\nu_i = |\kappa_\tau^2 Y_i| = \kappa_\tau^2 |Y_i|. \quad (34)$$

This is not yet a complete proof, because the hypercharge-directed connection $A_{\tau,i}(x) = \kappa_\tau^2 Y_i \tanh x$ must itself be derived from the parent metric, connection, index problem, or BRST quotient. It does, however, isolate the missing theorem: once the tau connection is fixed by the canonically normalized hypercharge generator, the zero-mode exponent follows algebraically.

This section should be read in the same spirit as a domain-wall zero-mode calculation: the zero mode follows from the first-order operator once the wall and the charge are known. The novelty claimed here is not the existence of a $\text{sech}'' x$ zero mode by itself. The novelty is the proposed route by which the $3 + 2$ stabilizer, hypercharge normalization, and wall quotient would select the exponent used in the Higgs overlap.

14 Stabilizer Origin Of The Hypercharge Direction

Branch A starts from the $3 + 2$ split

$$V = C \oplus W, \quad \dim C = 3, \quad \dim W = 2. \quad (35)$$

A connection that preserves the non-Abelian stabilizer $SU(3) \times SU(2)$ must be block-scalar on C and on W :

$$T = \text{diag}(a, a, a, b, b). \quad (36)$$

The traceless condition gives

$$3a + 2b = 0. \quad (37)$$

Thus, up to an overall normalization, there is only one Abelian generator that commutes with the Branch A non-Abelian stabilizer. Choosing the conventional Standard Model hypercharge signs gives

$$Y = \text{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right). \quad (38)$$

After canonical normalization this is the same direction as $-T_\Sigma$. Therefore a minimal stabilizer-preserving tau connection has no independent Abelian direction available other than hypercharge:

$$A_\tau(x) = f(x) T_Y. \quad (39)$$

This is the sense in which the hypercharge alignment is derived from the Branch A stabilizer. A stronger proof would still have to show that the parent connection dynamically chooses this minimal stabilizer-preserving Abelian direction rather than a larger symmetry-breaking sector.

15 Minimal Wall Profile And Quotient Vacua

Once the direction is fixed, the remaining question is the scalar interpolation $f(x)$. The minimal domain-wall ansatz assumes a smooth odd profile connecting two constant asymptotic tau vacua:

$$f(-\infty) = -1, \quad f(+\infty) = 1, \quad f(0) = 0. \quad (40)$$

The standard first-order kink equation

$$f'(x) = 1 - f(x)^2 \quad (41)$$

has the unique centered solution

$$f(x) = \tanh x. \quad (42)$$

Equivalently, this is the Bogomolny equation for the quartic domain-wall potential

$$V(f) = \frac{1}{2}(1 - f^2)^2, \quad (43)$$

after choosing the tau length scale so that the wall width is one. The same choice also yields the Pöschl–Teller supersymmetric zero-mode operator

$$Q_i = \partial_x + \kappa_\tau^2 Y_i \tanh x, \quad (44)$$

whose normalizable solutions are powers of $\text{sech } x$.

Thus the $\tanh x$ profile is not selected to fit the Higgs quartic. It is the minimal centered kink profile for a smooth two-vacuum tau wall. What remains open is whether the parent projection geometry or quotient action forces this minimal kink rather than allowing a more general monotone profile.

The two-vacuum part of this wall route can be given a sharper quotient interpretation. The Branch A Abelian direction is a one-dimensional internal line

$$L_Y = \text{span}(T_Y), \quad (45)$$

not necessarily a signed observable vector. If the parent regular background is the unoriented line, or equivalently the rank-one projector onto that line, then the two oriented representatives are identified:

$$\text{span}(T_Y) = \text{span}(-T_Y), \quad P_Y = P_{-Y}. \quad (46)$$

Thus the quotient identifies

$$\Sigma \sim -\Sigma. \quad (47)$$

For $\Sigma = fT_Y$, this gives

$$f \sim -f. \quad (48)$$

The regular invariants are sign-even:

$$\text{Tr}(\Sigma^2) = \frac{1}{2}f^2, \quad \text{Tr}(\Sigma'^2) = \frac{1}{2}(f')^2. \quad (49)$$

Therefore odd powers of f would not be well defined on the quotient. If the regular branch fixes the line magnitude,

$$2 \text{Tr}(\Sigma^2) = 1, \quad (50)$$

then $f^2 = 1$, giving the two oriented representatives $f = \pm 1$. In this sense, the two vacua are not two unrelated signed scalar states; they are the two orientations of the same internal Branch A line.

This closes the two-vacuum logic conditional on a parent theorem that the Branch A regular datum is an unoriented tau-normal line or projector. If the full parent theory instead treats $+T_Y$ and $-T_Y$ as distinct observable charges, this quotient argument fails and the two-vacuum wall must be supplied by a separate dynamical mechanism.

For the Higgs doublet, $Y_H = 1/2$, giving

$$\nu_D = \frac{3}{5} \cdot \frac{1}{2} = \frac{3}{10}. \quad (51)$$

The first-order operator is

$$Q_D = \partial_x + \frac{3}{10} \tanh x, \quad (52)$$

and the visible zero mode solves $Q_D h_D = 0$:

$$h_D(x) = \mathcal{N}_D \operatorname{sech}^{3/10} x. \quad (53)$$

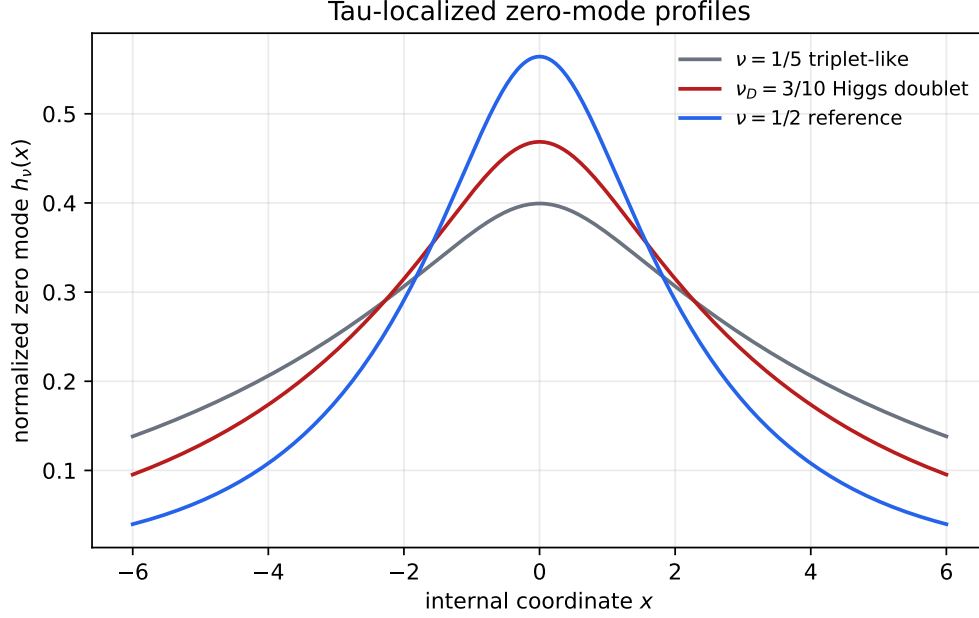


Figure 2: Normalized tau-localized zero-mode profiles. The Higgs doublet exponent $\nu_D = 3/10$ is compared with a triplet-like exponent and a reference exponent.

16 Quartic Overlap And Sensitivity Audit

For a normalized profile h_ν , the quartic localization functional is

$$I_4(\nu) = \int h_\nu^4 dx = \frac{\Gamma(\nu + 1/2)^2 \Gamma(2\nu)}{\sqrt{\pi} \Gamma(\nu)^2 \Gamma(2\nu + 1/2)}. \quad (54)$$

For $\nu_D = 3/10$ this gives

$$I_4(3/10) = 0.133756. \quad (55)$$

If the parent quartic is $\lambda_\tau (H^\dagger H)^2$, then

$$\lambda_H = \lambda_\tau I_4(3/10). \quad (56)$$

For $\lambda_H \simeq 0.129$, the implied parent quartic is

$$\lambda_\tau \simeq 0.964. \quad (57)$$

The important point is not the exact closeness to one. The safer statement is that the overlap requires an order-one parent quartic rather than an extreme hierarchy. The canonical parent quartic is not derived in this manuscript.

To reduce the numerology risk, the generator writes an explicit sensitivity table. In the moderate window $0.25 \leq \nu \leq 0.35$, the required parent quartic ranges approximately from 0.847 to 1.129. This means the mechanism is not a delta-function coincidence at $\nu = 0.3$: a neighborhood of nearby exponents still maps the observed quartic to an order-one parent coupling. The sensitivity audit strengthens the mechanism as a robust target, while still not proving the hypercharge-localization rule.

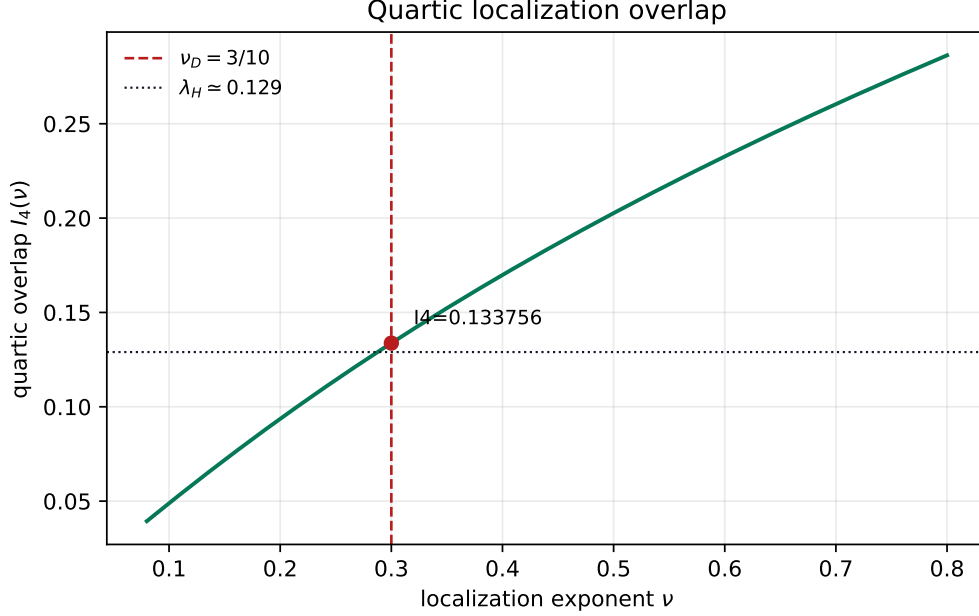


Figure 3: Quartic localization overlap as a function of the tau-localization exponent. The Branch A Higgs value $\nu_D = 3/10$ is marked, but the figure should be read as a sensitivity audit rather than as standalone evidence.

17 Cohomological Protection

The quotient $H \sim H + Q_D^\dagger \Lambda$ suggests a projection-BRST implementation,

$$sH = Q_D^\dagger c, \quad sc = 0, \quad s\bar{c} = b, \quad sb = 0. \quad (58)$$

At the exact cohomological point the critical operator is factorized,

$$\mathcal{O}_0 = Q_D^\dagger Q_D, \quad (59)$$

so the visible zero mode is massless. A local parent mass $\int H^\dagger H dx$ does not descend to the quotient because it is not invariant under the vertical redundancy. This is only a possible lightness-protection route. The current manuscript does not solve the Higgs hierarchy problem. A complete treatment would require the Hilbert-space domain, nilpotency, regulator, anomaly, and Ward-identity analysis. Here the projection-BRST language is retained as a roadmap gate, not as a completed QFT closure proof.

The role of this section is therefore structural. It states what kind of quotient would be needed for the light scalar to be protected. It does not replace a full gauge-theoretic BRST construction, and it does not replace the renormalized determinant calculation.

The missing QFT step is also specific. A completed version must construct the functional space on which Q_D acts, prove nilpotency after regularization, show that the quotient is anomaly-free, and derive Ward identities that forbid the dangerous scalar mass terms. Until those conditions are checked, the BRST language should be read as a protection criterion, not as protection already achieved.

18 Controlled Top/Flavor Deformation

The leading cohomology-breaking deformation is written as

$$\mathcal{O}_D = Q_D^\dagger Q_D - \delta_\star W_D, \quad W_D(x) = \frac{3}{10} \operatorname{sech}^2 x. \quad (60)$$

The zero-mode expectation value gives

$$\int h_D^2 W_D dx = \frac{9}{80}, \quad (61)$$

and therefore

$$\mu_H^2 = \frac{9}{80} \delta_\star M_\tau^2. \quad (62)$$

This section is a roadmap calculation: the top determinant must still be completed before the deformation can be promoted to a derived prediction. The current parent-hub refinement identifies the relevant top-sensitive slot as an up-type closure channel and records a same-domain wall- Q determinant pilot. These are gate refinements, not a completed determinant. In particular, the manuscript does not yet prove radiative stability or solve the hierarchy problem. The top/flavor determinant remains the hardest gate: it must show that mismatch-independent and linear mass terms are absent or quotient-trivial, and it must derive the physical Yukawa strength rather than inserting it.

This is where the proposal meets the usual naturalness problem most directly. The overlap calculation can make the tree-level quartic look natural, but it does not by itself control loop sensitivity. A successful completion must show that the same quotient/domain structure that produces the zero mode also controls the top-sensitive radiative terms. Otherwise the construction remains a localized-overlap model rather than a Higgs-naturalness mechanism.

The packet now includes a toy mass-derivative stress test for this gate. It computes the forbidden top-sensitive curvature proxy under paired and shifted positive spectra, and under two cutoff families. The paired toy spectrum stays zero under both regulators, while the shifted spectrum leaves a regulator-dependent residue. This is not the physical top determinant, but it fixes the diagnostic that the real calculation must pass.

19 What Is Reproduced And What Is Not

The current package reproduces:

- a minimal $3 + 2$ stabilizer-compatible hypercharge direction;
- conditional parent-selection gates linking two protected visible clusters, ϵ_3/ϵ_2 invariant roles, and the anomaly-safe target;
- a conditional F1–F8 localization-forcing route for $\nu_i = \kappa_\tau^2 |Y_i|$;

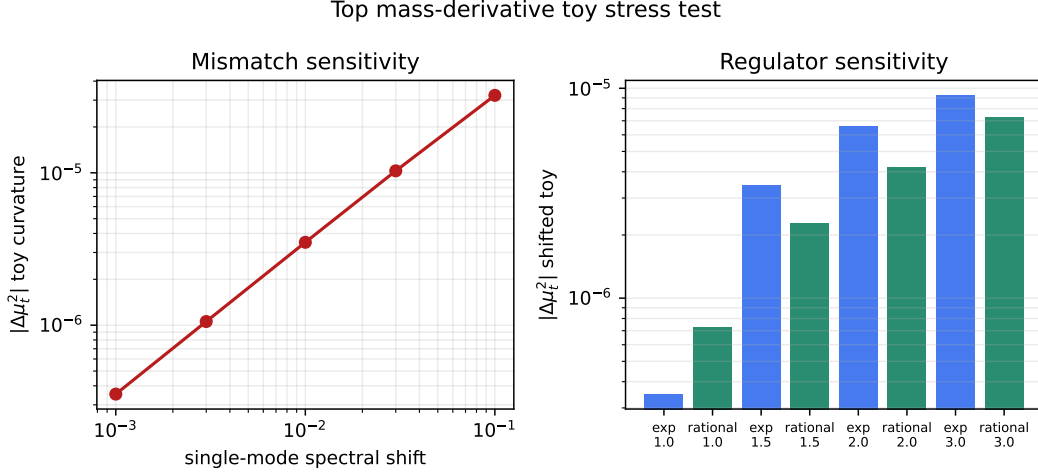


Figure 4: Toy mass-derivative stress test for the top determinant gate. The figure illustrates the required diagnostic behavior: same-domain paired spectra remove the forbidden curvature, while a small spectral/domain mismatch leaves a regulator-sensitive residue.

- the Branch A working exponent $\nu_D = 3/10$ once the $\nu_i = 3|Y_i|/5$ rule is assumed;
- the normalized $\text{sech}^{3/10}$ zero mode;
- the quartic overlap $I_4(3/10) = 0.133756$;
- an order-one parent-quartic requirement and a TeV-scale deformation estimate;
- a sharpened statement of where the wall- Q and top/Yukawa gates must enter.
- a v0.2 single-package scaffold joining compact cell, wall domain, same-domain regulator/product bookkeeping, and the ϵ_τ scale gate;
- the fixed dimensionless coefficient $y_t^{\text{parent}} = 0.9615522319$ inside that scaffold, before absolute-scale and matching closure.

It does not reproduce:

- the full Standard Model;
- the Yukawa hierarchy;
- the full parent-action derivation of the F1–F8 forcing conditions;
- anomaly cancellation from a derived parent representation construction;
- derivation of the 3×2 bridge and singlet/color channels from the parent action;
- radiative stability;
- the measured Higgs mass from a completed top determinant;
- the top Yukawa strength or family hierarchy;

- the absolute electroweak scale, because ϵ_τ , C_A , and the compact-cell/wall embedding are not yet derived from the final parent action;
- a collider-ready heavy-sector spectrum.

20 Wolfram Language Audit

The repository includes optional Wolfram Language audit scripts as an independent symbolic/numeric check of the formal skeleton:

- `Higgs_Quartic_Overlap_Verification.wl` checks the normalized $\text{sech}'' x$ profile, the quartic-overlap formula, $I_4(3/10)$, and the sensitivity range for the required parent quartic;
- `BranchA_Stabilizer_Hypercharge_Audit.wl` checks $\text{Tr } T_\Sigma = 0$, $\text{Tr } T_\Sigma^2 = 1/2$, $\text{Tr } T_Y^2 = 1/2$, and $T_\Sigma = -T_Y$, thereby verifying the stabilizer and hypercharge-normalization origin of the 3/5 factor;
- `G2_Unoriented_Line_Quotient_Audit.wl` checks that the line projector for T_Y is invariant under $T_Y \mapsto -T_Y$, that $\text{Tr}[(fT_Y)^2] = f^2/2$, and that the unit-line condition gives the two oriented representatives $f = \pm 1$;
- `Projection_BRST_Skeleton.wl` checks only the algebraic skeleton $s^2 H = 0$ and $Q_D h_D = 0$.
- `Compact_Gate_Ledger_Audit.wl` checks that the generated full-derivation ledger has continuous C0–C20 gate IDs, includes the C16/C17/C20 subgates, records the finite-residue and endpoint-matching gates, and keeps the claim boundary at proposition/program level rather than solved-proof level.

The generated log files are included in the reproducibility packet. These checks support the formula and algebra audit. They verify the normalization source of 3/5, but they do not derive the localization postulate $\nu_i = \kappa_\tau^2 |Y_i|$, prove anomaly freedom, prove regulator safety, or establish radiative stability.

21 Why This Is Not Just Numerology

The quartic-overlap calculation is evidence for internal coherence of the Branch A Higgs module, but not evidence for the full physical theory. The module becomes scientifically decisive only if the gates are passed in the right order: first derive the localization postulate that ties the exponent to $\kappa_\tau^2 |Y_i|$, then prove the quotient/anomaly safety, then complete the top determinant, and only then compare the resulting heavy-sector predictions to collider constraints. Without those gates, the value $I_4(3/10)$ should be treated as a reproducible mechanism target rather than as physical validation.

22 What Would Count As A Higgs-Sector Proof

The proof target is stronger than the scaffold constructed in this manuscript. It is the existence of a final parent action

$$S_{\text{parent}}[\Psi] \implies \text{Branch A gates} \implies \text{SM-compatible Higgs sector}, \quad (63)$$

where the implication is dynamical rather than retrospective. In particular, S_{parent} would have to select the compact tau cell, the two protected visible modules, the $3 + 2$ carrier, the hypercharge wall line, the BPS wall profile, the same-domain Q /regulator package, and the scale unit ϵ_τ without using the measured Higgs mass, Higgs vev, or top mass.

Only after that would the remaining QFT tests become meaningful: anomaly closure, regulator independence, Ward identities, top-determinant cancellation, running/matching to the electroweak scale, and collider-compatible heavy-sector predictions. Thus the present result is not a Higgs proof. It is a paper-level reduction of the proof problem to a small number of explicitly named gates.

23 Remaining Proof Gates

The present manuscript isolates the unresolved gates. These are not presentation details; they are the conditions that separate the current mechanism target from a Higgs-sector derivation.

1. **Explicit parent action and UV/continuum completion.** The v0.2 single-package action is a constrained scaffold, not the final microscopic theory. A proof must define the parent configuration space, dynamical fields, symmetry principle, measure, continuum limit, and admissible projection map without using Higgs or top endpoints as inputs.
2. **Localization derivation.** The factor $3/5$ is now traced to the Branch A hypercharge normalization, and the localization rule has been sharpened into a conditional F1–F8 forcing route. A completed proof must still derive those F-conditions from the parent action: the two protected visible modules, the unoriented hypercharge line/projector, the absence of non-hypercharge leakage, the minimal BPS two-vacuum functional, and the same-domain Hessian/ Q factorization.
3. **Parent-selection and representation derivation.** The current packet narrows the $3 + 2$ carrier to a conditional chain of two protected visible clusters, minimal $3 + 2$ support, ϵ_3/ϵ_2 invariant roles, and an invariant/anomaly bridge. A proof must still derive those roles, the 3×2 bridge, and the singlet/color channels from the parent cohomology or Hessian structure rather than declaring an SM-like target.
4. **BRST/anomaly/regulator consistency.** The projection-BRST skeleton explains how a quotient could protect the visible zero mode, but the present packet checks only the algebraic skeleton. A proof must still define the Hilbert-space domain, nilpotent charge, regulator, anomaly constraints, and Ward identities. It must also show that these identities survive the same continuum/UV completion used for the top determinant.
5. **Top determinant and radiative stability.** The top/flavor deformation is still a roadmap calculation. The up-type top-sensitive slot and same-domain wall- Q pilot sharpen where the calculation must occur, but they do not replace it. Until the physical determinant is computed, the Yukawa strength is derived, and mismatch-independent or linear mass terms are shown to be absent or quotient-trivial, the manuscript does not solve the Higgs hierarchy problem. This is the decisive naturalness gate: a localized tree-level quartic is not enough unless the radiative scalar-mass sensitivity is also controlled.
6. **Compact tau-cell and scale.** The v0.2 single-package scaffold isolates the remaining absolute unit as ϵ_τ inside $\Lambda_A = C_A \epsilon_\tau \mu_\tau(\mathcal{K}_\tau) N_\tau \hat{E}_{\text{wall}}$. A proof must still specify the geometry/topology/spectral structure of $\mathcal{K}_\tau$, derive this universal parent density or stress-energy

scale, derive C_A , and show that the Branch A wall domain is induced from the same compact tau cell rather than declared as a disconnected Higgs-sector input. The most useful positive result would be a compact spectral or cell geometry whose low-lying sector selects the $3 + 2$ stabilizer and whose defect sector induces the hypercharge wall.

7. **Explicit compact-geometry v0.1.** The current v0.1 target is the orbifold spectral package $\mathcal{K}_\tau = S^1/\mathbb{Z}_2$ with algebra $(C^\infty(\mathcal{K}_\tau) \otimes (M_3(\mathbb{C}) \oplus M_2(\mathbb{C})))^{\mathbb{Z}_2}$, Hilbert domain $L^2(\mathcal{K}_\tau) \otimes (\mathbb{C}^3 \oplus \mathbb{C}^2) \otimes \mathbb{C}_{\text{wall}}^2$, and Dirac-like wall operator $D_\tau = \sigma_1 \partial_x + \sigma_2 \kappa_\tau^2 T_Y f(x) + D_{\text{gap}}$. A proof must still show that the self-adjoint orbifold domain, index residue, positive-spectrum pairing, and regulator all come from this single package.
8. **Standard Model emergence.** A broader Branch A emergence program must still derive the representation content, gauge dynamics, anomaly cancellation, Yukawa pattern, and low-energy effective field theory from the parent action. This paper contributes only a Higgs-module candidate gate.

These gates make the status of the paper precise. The quartic-overlap result is a reproducible and nontrivial mechanism target, but the localization, cohomological consistency, and radiative-stability gates must all close before the module can be promoted to a derivation.

24 Compact Spectral Protection Program

The technical companion keeps the full operator-domain ledger, but the main paper needs the core logic explicitly. The protection route is not merely a localized wave-function ansatz. It is the following compact spectral program:

$$\text{compact domain} \rightarrow Q\text{-paired self-adjoint operators} \rightarrow \text{index residue} \rightarrow \text{zeta-determinant cancellation} \rightarrow \text{allowed f} \quad (64)$$

The candidate compact geometry is the orbifold package $\mathcal{K}_\tau = S^1/\mathbb{Z}_2$ with a first-order wall operator Q_ν . The required domain theorem is:

$$Q_\nu : \mathcal{D}_- \rightarrow \mathcal{D}_+, \quad H_- = Q_\nu^\dagger Q_\nu, \quad H_+ = Q_\nu Q_\nu^\dagger, \quad (65)$$

with $H_\pm$ self-adjoint on one shared compact domain. If this holds, then for every positive eigenvalue $\lambda > 0$ the map

$$\psi \mapsto \lambda^{-1/2} Q_\nu \psi \quad (66)$$

pairs the positive spectra of H_- and H_+ with equal multiplicity. The bulk positive-mode determinant then cancels at the zeta level, leaving only kernel/index, anomaly/Ward, boundary, or frozen matching residues.

The negative status is equally important:

- a conditional compact spectral theorem is established only within the stated operator-domain assumptions; the parent action has not yet selected the required compact domain;
- conditional positive-spectrum zeta-determinant cancellation follows within the same-domain setup, and a physical top finite-residue extraction gate is formulated, but the finite residue has not yet been computed from the parent QFT;
- same-domain anomaly/Ward and representation trace-cancellation gates are formulated, but the parent-derived representation complex has not yet been computed;

- a UV/continuum admissibility criterion and microscopic completion target are formulated, but no convergent parent action family is supplied;
- a physical matching map and endpoint protocol are formulated, but no numerical matching to the measured Higgs vev, top mass, or collider observables has been performed.

This is why the compact spectral program is central to the paper. Without it, the construction is only a localization model. With it, the Higgs-lightness question becomes a sharply defined spectral problem: prove the self-adjoint domain, prove positive-spectrum pairing, prove regulator/Ward compatibility, and then compute the top-sensitive finite residue from the parent operator family.

25 Next Theorem Handoff

The related structures in Section 2 are useful only if they sharpen the next calculations. The immediate handoff is:

Open theorem	Best external toolkit	Concrete next deliverable
protected-form selection	cohomology and index theory	define the closure and pairing complexes, prove which low-degree residues vanish, and isolate H_{cl}^3 and H_{pair}^2 as first protected classes
compact spectrum	spectral geometry / self-adjoint extension theory	replace the finite-box C9 pilot by a $\mathbb{Q}$ -paired compact $S^1/\mathbb{Z}_2$ domain and compute or bound the spectrum
determinant finiteness	zeta determinant and heat-kernel methods	prove that the same-domain positive spectra cancel up to zero-mode/index residue
quotient consistency	BRST cohomology and Ward identities	construct the nilpotent charge, physical cohomology, anomaly test, and Ward identities on the same domain
radiative stability	effective QFT matching	compute the top-sensitive determinant and show whether mismatch-independent scalar mass terms are absent or quotient-trivial

This table is intentionally operational. A future paper can close one row without claiming the whole Higgs sector. The strongest next target is the protected-form selection row, because it would turn the $3 + 2$ carrier from a minimal candidate into a parent-selected algebraic consequence.

26 Claim-Upgrade Ladder

The manuscript should be read with the following claim ladder. Each level licenses only the next stronger statement after the listed gate is closed.

Closed gate	Allowed claim	Still forbidden
quartic overlap only	reproducible mechanism target	Higgs derivation or naturalness solution
localization rule derived	Branch A maps Higgs quartic to an order-one parent coupling	Standard Model derivation
protected-form/index selection	3 + 2 carrier becomes parent-selected rather than postulated	full gauge/Yukawa sector
shared compact spectrum and C18 pairing	positive-spectrum cancellation target becomes theorem-level	physical radiative stability
C19–C20 determinant residue admitted	determinant route becomes regulator-safe candidate	measured Higgs/top masses
top determinant and matching closed	Higgs-lightness mechanism can be claimed within Branch A	full Tau Core or TOE validation

At the present stage the paper sits at the first level and partially organizes the next levels. It does not yet license the later claims.

27 Near-Term Falsifiable Prediction

The current module predicts a narrow kind of future test rather than an already validated signal. If the top/flavor deformation gate is completed with a natural spurion $\delta_\star \sim 10^{-3}$, the same equations imply a parent scale M_τ in the multi-TeV range and a Higgs-sector spectral threshold

$$m_{\text{gap}} \simeq \nu_D M_\tau \sim 2\text{--}4 \text{ TeV}. \quad (67)$$

The associated Higgs-coupling deviations should be parametrically of order

$$\frac{\Delta g}{g} \sim \frac{v^2}{m_{\text{gap}}^2} \sim 10^{-3}\text{--}10^{-2}. \quad (68)$$

This is a falsifiable target, not a confirmed prediction. The module weakens if the completed determinant forces much larger deviations, no TeV-sector threshold, or collider-excluded states.

28 Validation Gates

The module would fail if any of the following gates fail:

- the Branch A projection metric rule $\nu_i = 3|Y_i|/5$ cannot be forced from the parent action or the F1–F8 sufficient conditions fail;
- the two-block / 3 + 2 / invariant-role chain cannot be derived from the parent action or stability structure;
- the 3×2 bridge and anomaly-safe visible target cannot be obtained without post-hoc representation choice;
- the physical Q_μ operator cannot be derived from the same wall-Hessian/domain package;
- the parent quartic is not canonical or requires large tuning;

- the projection-BRST quotient is anomalous or regulator-dependent;
- a visible Higgs mass is unavoidable at $\delta_\star = 0$;
- the top determinant produces a large mismatch-independent or linear mass term, or requires an inserted Yukawa hierarchy;
- ϵ_τ , C_A , or the wall-cell embedding must be chosen from the measured Higgs vev or top mass;
- the TeV-sector spectral window is excluded by Higgs coupling or direct-search constraints.

29 Observer realization: inherited results and physical boundary

The current observer programme distinguishes a conditional mathematical construction from an occupied physical measuring system. The source order remains base plus one universal seed, stabilized morphological response, observer–source restriction, shared descriptor, and resolved terminal readouts. Neither a detector nor an empirical residual constructs the parent body backwards. Dynamics below concern downstream subsystems only.

The canonical observer-origin derivation, Sections 22–26, owns the following results. The existing joint source action fixes static and frequency-dependent response together. Variation of the BRAC potential couples to the field square; a linear field coupling is a distinct comparator unless derived by expansion about an occupied background. Fixed canonical means select a unique coherent state in a finite positive harmonic packet only if there is no excess fluctuation energy. This is a free/asymptotic preparation statement, not the ground state of an interacting detector. For the existing coupled harmonic pair, the joint ground is pure but either local subsystem is mixed, with its covariance fixed by the complete joint Hamiltonian.

These are conditional derivations and internal numerical checks. The actual observer–source context, physical preparation process, kinetic/energy calibration and stable operational resolution remain unproved. The existing ORA-T1 selector conditionally chooses stable quantizers when the occupied context, admissible partition family and positive closure barrier are supplied; it does not establish those physical premises. A computed ground state is not a preparation mechanism, and a covariance or optimal abstract measurement is not an implemented stable quantizer Q_{OS} .

Ownership and use. The central observer-origin note owns the new contact, state and covariance derivations; the body-readout technical companion carries the consolidated mathematics. Foundation Papers IV, V and VI retain physical-selection, clock-descent and quantum-readout ownership, respectively. This paper imports only the consequences within its scope. No observational dataset, baseline, frozen coefficient, score or empirical conclusion is revised by this update. The reproducibility supplement records the source hashes and the exact conditional/physical distinction.

Consequence for this paper. These results do not select a particle spectrum, coupling constant, Higgs vacuum or gravitational terminal. Common source ancestry remains weaker than physical realization of those sectors.

30 Source identification and recovery

The current observer audit separates a conditional forward construction from physical source selection. A frozen body, occupied observer incidence, preparation, complete instrument and stable

resolution can determine records without selecting those inputs from Nature. Equal effects do not fix postmeasurement states, and symmetry within a chosen interaction family does not exclude other symmetric couplings. These results do not select a particle spectrum, coupling constant, Higgs vacuum or gravitational terminal. Common source ancestry remains weaker than physical realization of those sectors.

Dependency and claim ownership. Foundation VI and the body-readout technical companion own the finite instrument/access proof; Foundation IV and TD-0 retain physical source selection; Foundation V, the temporal synthesis and seed-body companion own the gradient/inertial distinction. This paper inherits only the stated specialization. Canonical assumptions, countermodels and calibration audits are in https://github.com/tau-core-research/tau-core-theory/blob/main/docs/tau_rqm_instrument_to_record_bridge_001.md. No physical seed, Nature occupation or Tau-exclusive observation is established by this update. Existing empirical endpoint definitions and scores are unchanged.

31 Conditional source preparation and time-resolved access

The common-operator scalar-Dirac completion now supports an explicit conditional chain from interaction vertices to pair transfer and joint readout. Nonzero vertices do not select occupation. Relational coherence can be generated from an occupied number-diagonal pair even while both local mean fields vanish. In a declared resonant rotating-wave and Markov loss model, finite-window emission retains this coherence information, whereas an all-time count discards it. This is standard-compatible quantum recovery, not a Tau-specific observational residual. These results do not select a particle spectrum, coupling constant, Higgs vacuum or gravitational terminal. Common source ancestry remains weaker than physical realization of those sectors.

Frozen model and preparation boundary. The conditional completion uses a constant scalar kinetic metric K , a stationary potential Hessian H_* , and a common Dirac mass operator gD_* . Scalar masses obey $H_*u = M^2Ku$ and fermion masses are $g|d_i|$. The signed masses are retained in the transformed Yukawa spin factors. For the four selected pump directions, the inclusive fermionic partial widths at the stipulated $g = 1$ benchmark are 0.10031707, 0.12080077, 0.15789991, 0.14826014 in model mass units. Their ratios to scalar mass range from 0.09459 to 0.11929; these are not measured lifetimes or a parametrically narrow-resonance proof. Single-fermion emission into these pumps is kinematically closed at this point; inverse fermion-antifermion preparation has allowed on-shell momenta, but no incoming population is thereby selected.

For a canonical homogeneous mode in a declared volume, a displacement $A\sqrt{w_C}$ with zero mean velocity has minimum coherent density $n_C = M_C A^2 w_C / 2$. Positivity gives $\langle N \rangle \geq |\langle a \rangle|^2$; a displaced thermal state has the same mean and larger occupation. This is a conditional bound, not selection of a coherent initial state.

Existing interaction and relational information. On an isolated resonant pair/scalar sector let $B = |\text{empty}\rangle\langle\text{pair}|$ and $H_I = Ga^\dagger B + G^* B^\dagger a$. Choose a real positive G convention and $\theta = Gt$. An initially occupied pair probability p with no local phase coherence evolves to

$$\langle N_\phi \rangle = p \sin^2 \theta, \quad \langle a \rangle = 0, \quad \langle a^\dagger B \rangle = ip \sin \theta \cos \theta.$$

The last quantity is invariant under common total-number phase rotation. It requires occupation, not an external absolute phase reference. A single pair nevertheless cannot prepare a nonzero exact

coherent oscillator state. With ideal number detection after a continuation angle δ , on the basis $|P\rangle, |S\rangle$,

$$E_\delta = \frac{1}{2}[I - \cos(2\delta)Z - \sin(2\delta)Y].$$

Three settings determine both populations and the imaginary joint coherence, but the real coherence is an exact null of this measurement family. This does not permit deleting that direction from other readouts.

Simultaneous exchange and loss. For a separately justified constant-rate Markov reservoir, distinct from the coherent pair, the no-emission operator and finite-window effect are

$$H_{\text{eff}} = \begin{pmatrix} 0 & G \\ G & -i\Gamma/2 \end{pmatrix}, \quad K_0(T) = e^{-iH_{\text{eff}}T}, \quad E_{\text{em}}(T) = I - K_0(T)^\dagger K_0(T).$$

Differentiation gives

$$E_{\text{em}}(T) = \int_0^T \Gamma K_0(t)^\dagger |S\rangle\langle S| K_0(t) dt, \quad 0 \leq E_{\text{em}} \leq I.$$

At the declared control $\Gamma/G = GT = 1$, the joint state with diagonal entries $1/2$ and off-diagonal entry $i/2$ has emission probability 0.5581543931 ; its dephased control has 0.3385743227 . Independent quadrature and matrix exponentiation agree to below 6×10^{-16} over twenty tested windows. These are internal controls, not measurements. For $G \neq 0$ and $\Gamma > 0$, $E_{\text{em}}(\infty) = I$ on the occupied sector: all-time counting discards the distinguishing coherence. Finite timing resolution is therefore essential for this particular signal.

What is not closed. Emission is not a registered or amplified record. A collection fraction multiplies the effect; unknown efficiency and occupation remain degenerate. The normalized coherent coupling G is not the inclusive continuum width Γ , and neither fixes SI seconds. Physical preparation, mode normalization, reservoir approximation, gauge completion, detector and clock calibration, and stable Q_{OS} remain unselected. Standard quantum exchange reproduces these conditional results; no new empirical score or Tau-exclusive prediction follows.

Dependency ownership and limits. The P4 common-source assembly owns the scalar/Yukawa benchmark and finite preparation calculations; Foundation VI and the body-readout companion inherit the joint-access mathematics. Foundation IV and TD-0 retain physical source selection, and Foundation V retains clock descent. The parent remains atemporal: the evolution above is a conditional downstream 4D model, not parent memory or a derived morphogenetic time. Physical initial occupation, a gauge-complete realization, detector amplification, absolute calibration and stable observer-relative Q_{OS} remain unselected. No empirical endpoint or score is changed. The canonical derivation and reproducible numerical sources are linked at https://github.com/tau-core-research/tau-core-theory/blob/main/docs/tau_core_p4_nonabelian_particle_com_mon_source_assembly_001.md.

32 Conclusion

The Branch A Higgs module links a $3 + 2$ stabilizer, hypercharge normalization, a $\text{sech}^{3/10}$ visible zero mode, and a quartic overlap requiring an order-one parent quartic. The paper establishes a compact, reproducible mechanism target: if the Branch A localization rule is derived, the Higgs

quartic is mapped to a natural parent-scale coupling rather than an extreme hierarchy. The hard blockers are now sharply identified: an explicit final parent action, continuum/UV completion, a concrete compact tau-cell geometry, QFT-level BRST/anomaly/regulator closure, top-determinant radiative stability, and physical matching. The manuscript still does not establish the full parent theory, the measured Higgs vev, the measured top mass, the Standard Model, or the Higgs hierarchy solution. The present result is a mechanism target and spectral proof program, not a completed Higgs-sector derivation.

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